

Optimal Control
with
Aerospace Applications

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Chapter 10

General Theory of Optimal Rocket Trajectories

10.1 Introduction

In this chapter we develop a general theory of optimal spacecraft trajectories based on two pioneering works: Breakwell [1959] and Lawden [1963]. Lawden introduced the concept of the primer vector, which plays a dominant role in minimum-propellant trajectories and also in other types of optimal trajectories. A more complete discussion of the topics in this chapter, including several example trajectories, is in Prussing [2010].

10.2 Equations of Motion

The equation of motion of a spacecraft which is thrusting in a gravitational field can be expressed in terms of the orbital radius vector $\mathbf{r}$ as:

$$\ddot{\mathbf{r}}(t) = \mathbf{g}(\mathbf{r}) + \mathbf{\Gamma}(t), \quad \mathbf{\Gamma}(t) = \Gamma(t)\mathbf{u}(t) \quad (10.1)$$

The variable $\mathbf{\Gamma}$ is the thrust acceleration vector, whose magnitude Γ is defined as the thrust (force), T , divided by the mass of the vehicle, m . The variable $\mathbf{u}$ is a unit vector in the thrust direction, and $\mathbf{g}(\mathbf{r})$ is the gravitational acceleration vector. A careful derivation of Eq. (10.1) requires deriving the *rocket equation* by equating the net external force (such as gravity) to the time rate of change of the linear momentum of the vehicle/exhaust particle system (see sections 5.1–5.4 of Prussing and Conway [2013]).

An additional equation expresses the change in mass of the spacecraft due to the generation of thrust:

$$\dot{m} = -b, \quad b \geq 0 \quad (10.2)$$

In Eq. (10.2) b is the (nonnegative) mass flow rate. The thrust magnitude, T , is given by $T = bc$, where c is the *effective* exhaust velocity of the engine. The word “effective”

applies to high-thrust chemical engines where the exhaust gases may not be fully expanded at the nozzle exit. In this case there exists an additional contribution to the thrust which is incorporated into the effective exhaust velocity by defining

$$c \equiv c_a + (p_e - p_\infty) \frac{A_e}{b} \quad (10.3)$$

In Eq. (10.3) the subscript “e” refers to the pressure and area at the nozzle exit, c_a is the actual exhaust velocity at the exit, and p_∞ is the ambient pressure. If the gases are exhausted into the vacuum of space, $p_\infty = 0$.

An alternative to specifying the effective exhaust velocity is to describe the engine in terms of its *specific impulse*, defined to be:

$$I_{sp} \equiv \frac{(bc)\Delta t}{(b\Delta t)g_o} = \frac{c}{g_o} \quad (10.4)$$

where g_o is the standard acceleration of free fall on Earth, equal to 9.80665 m/s^2 . As shown in Eq. (10.4) the specific impulse is obtained by dividing the mechanical impulse delivered to the vehicle by the weight of propellant consumed. The mechanical impulse provided by the thrust force over a time Δt is simply $(bc)\Delta t$ and, in the absence of other forces acting on the vehicle, is equal to the change in its linear momentum. The weight (on Earth) of propellant consumed during that same time interval is $(b\Delta t)g_o$. We note that if instead one divides by the *mass* of the propellant (which, of course, is the fundamental measure of the amount of substance), the specific impulse would be identical to the exhaust velocity. However, the definition in Eq. (10.4) is in standard use with the value typically expressed in units of seconds.

10.3 High and Low-Thrust Engines

We can distinguish between high- and low-thrust engines based on the value of the nondimensional ratio Γ_{max} / g_o . For high-thrust devices this ratio is greater than unity and thus these engines can be used to launch vehicles from the surface of the Earth. This ratio may extend to as high as 100. The corresponding range of specific impulse values is between 200 and approximately 850 s, with the lower values corresponding to chemical rockets, both solid and liquid, and the higher values corresponding to nuclear thermal rockets.

For low-thrust devices the ratio Γ_{max} / g_o is quite small, ranging from approximately 10^{-2} down to 10^{-5} . These values are typical of electric rocket engines such as magnetohydrodynamic (MHD), plasma arc, and ion engines, and also for solar sails. The ratio for solar sails is of the order of 10^{-5} . An electric engine requires a separate power generator such as a radioisotope thermoelectric generator (RTG) or solar cells.

Other engine designs such as the magnetoplasmadynamic (MPD) thrusters and the variable specific impulse magnetoplasma rocket (VASIMR) seek to achieve higher levels of thrust while maintaining the high specific impulse typical of low-thrust engines.

10.4 Cost Functionals for Rocket Engines

Two basic types of rocket engines exist: *constant specific impulse* (CSI) and *variable specific impulse* (VSI), also called *power limited* (PL) engines. The CSI category includes both high- and low-thrust devices. The mass flow rate b in some cases can be continuously varied, but is limited by a maximum value b_{max} . For this reason a CSI engine is also described as a thrust-limited engine, with $0 \leq \Gamma \leq \Gamma_{max}$.

The VSI category includes those low-thrust engines, such as electric engines, which need a separate power source to run the engine. For these engines, the power is limited by a maximum value P_{max} , but the specific impulse can be varied over a range of values. The propellant expenditure for the CSI and VSI categories is handled separately.

The equation of motion, Eq. (10.1), can be expressed as:

$$\dot{\mathbf{v}} = \frac{cb}{m} \mathbf{u} + \mathbf{g}(\mathbf{r}), \quad \frac{cb}{m} \equiv \Gamma \quad (10.5)$$

For the CSI case we solve Eq. (10.5) using the fact that c is constant as follows:

$$d\mathbf{v} = \frac{cb}{m} \mathbf{u} dt + \mathbf{g}(\mathbf{r}) dt \quad (10.6)$$

Using Eq. (10.2),

$$d\mathbf{v} = -c\mathbf{u} \frac{dm}{m} + \mathbf{g}(\mathbf{r}) dt \quad (10.7)$$

which can be integrated (assuming constant $\mathbf{u}$) to yield:

$$\Delta \mathbf{v} = \mathbf{v}(t_f) - \mathbf{v}(t_o) = -c\mathbf{u} (\ln m_f - \ln m_o) + \int_{t_o}^{t_f} \mathbf{g}(\mathbf{r}) dt \quad (10.8)$$

or

$$\Delta \mathbf{v} = c\mathbf{u} \ln \left(\frac{m_o}{m_f} \right) + \int_{t_o}^{t_f} \mathbf{g}(\mathbf{r}) dt \quad (10.9)$$

which correctly indicates that, in the absence of gravity, the velocity change would be in the thrust direction $\mathbf{u}$. The actual velocity change achieved also depends on the gravitational acceleration $\mathbf{g}(\mathbf{r})$ which is acting during the thrust period. The term in Eq. (10.9) involving the gravitational acceleration $\mathbf{g}(\mathbf{r})$ is called the *gravity loss*. We note that there is no gravity loss due to an (instantaneous) impulsive thrust, described in section 10.5.2.

If we ignore the gravity loss term for the time being, a cost functional representing propellant consumed can be formulated. As will be seen, minimizing this cost functional is equivalent to maximizing the final mass of the vehicle. Since the thrust is equal to the product of the mass flow rate b and the exhaust velocity c , we can write:

$$\dot{m} = -b = \frac{-m\Gamma}{c} \quad (10.10)$$

$$\frac{dm}{m} = -\frac{\Gamma}{c} dt \quad (10.11)$$

For the CSI case the exhaust velocity, c , is constant and Eq. (10.11) can be integrated to yield

$$\ln \left(\frac{m_f}{m_o} \right) = -\frac{1}{c} \int_{t_o}^{t_f} \Gamma dt \quad (10.12)$$

or

$$c \ln \left(\frac{m_o}{m_f} \right) = \int_{t_o}^{t_f} \Gamma dt \equiv J_{\text{CSI}} \quad (10.13)$$

J_{CSI} is referred to as the *characteristic velocity* of the maneuver or the $\Delta \mathbf{V}$ (pronounced “delta vee”) and it is clear from Eq. (10.13) that minimizing J_{CSI} is equivalent to maximizing the final mass m_f . This form for J_{CSI} is also derived in Marec [1979].

In the impulsive thrust approximation for the unbounded thrust case ($\Gamma_{\text{max}} \rightarrow \infty$) the vector thrust acceleration is represented by

$$\mathbf{\Gamma}(t) = \sum_{k=1}^n \Delta \mathbf{v}_k \delta(t - t_k) \quad (10.14)$$

with $t_o \leq t_1 < t_2 \dots < t_n \leq t_f$ representing the times of the n thrust impulses. (See sections 6.1–6.3 of Prussing and Conway [2013].) Using the definition of a unit impulse,

$$\int_{t_k^-}^{t_k^+} \delta(t - t_k) dt = 1 \quad (10.15)$$

where $t_k^\pm \equiv \lim_{\epsilon \rightarrow 0} (t_k \pm \epsilon)$, $\epsilon > 0$.

Using Eq. (10.14) in Eq. (10.13) we obtain:

$$J_{\text{CSI}} = \int_{t_o}^{t_f} \Gamma dt = \sum_{k=1}^n \Delta v_k \quad (10.16)$$

and the total propellant cost is given by the sum of the magnitudes of the velocity changes.

The corresponding cost functional for the VSI case is obtained differently. The exhaust power (stream or beam power) is half of the product of the thrust and the exhaust velocity:

$$P = \frac{1}{2} T c = \frac{1}{2} m \Gamma c = \frac{1}{2} b c^2 \quad (10.17)$$

Using Eq. (10.17) along with

$$\frac{b}{m^2} = \frac{-\dot{m}}{m^2} = \frac{d}{dt} \left(\frac{1}{m} \right) \quad (10.18)$$

results in

$$\frac{d}{dt} \left(\frac{1}{m} \right) = \frac{\Gamma^2}{2P} \quad (10.19)$$

which integrates to

$$\frac{1}{m_f} - \frac{1}{m_o} = \frac{1}{2} \int_{t_o}^{t_f} \frac{\Gamma^2}{P} dt \quad (10.20)$$

Maximizing m_f for a given value of m_o regardless of whether it is optimal or not is obtained by running the engine at maximum power $P = P_{max}$. This conclusion is not as obvious as it looks in Eq. (10.20), because the value of Γ might be different for different values of P . To see that the engine should be run at maximum power we note that for a specified trajectory, $\mathbf{r}(t)$, the required vector thrust acceleration is given by Eq. (10.1) as

$$\mathbf{\Gamma}(t) = \ddot{\mathbf{r}}(t) - \mathbf{g}[\mathbf{r}(t)] \quad (10.21)$$

Thus, for a given trajectory $\mathbf{r}(t)$ (optimal or not), the final mass in Eq. (10.20) is maximized by running the engine at maximum power.

For this reason the VSI cost functional can be taken to be

$$J_{\text{VSI}} = \frac{1}{2} \int_{t_o}^{t_f} \Gamma^2 dt \quad (10.22)$$

This form for J_{VSI} is also derived in Marec [1979].

To summarize, the cost functionals representing minimum-propellant expenditure are given by

$$J_{\text{CSI}} = \int_{t_o}^{t_f} \Gamma dt \quad (10.23)$$

and

$$J_{\text{VSI}} = \frac{1}{2} \int_{t_o}^{t_f} \Gamma^2 dt \quad (10.24)$$

We see from Eqs. (10.23) and (10.24) that the minimum-propellant cost can be written in terms of the control magnitude $\Gamma(t)$ rather than introducing the mass as an additional state variable whose final value is to be maximized.

10.5 First-Order Necessary Conditions

10.5.1 Optimal Constant Specific Impulse Trajectory

For a constant specific impulse (CSI) engine the thrust is bounded by $0 \leq T \leq T_{max}$ (where T_{max} is a constant), corresponding to bounds on the mass flow rate: $0 \leq b \leq b_{max}$

(where b_{max} is a constant). Note that we can also prescribe bounds on the thrust acceleration (thrust per unit mass) $\Gamma \equiv T/m$ as $0 \leq \Gamma \leq \Gamma_{max}$, where Γ_{max} is achieved by running the engine at T_{max} . However, Γ_{max} is not constant, but increases due to the decreasing mass. One must keep track of the changing mass in order to compute Γ for a given thrust level, but this is easy to do, especially if the thrust is held constant, e.g., at its maximum value. However, if the propellant mass required is a small fraction of the total mass, a constant Γ_{max} approximation can be made.

The cost functional representing minimum-propellant consumption for the CSI case is given in Eq. (10.13) as

$$J = \int_{t_o}^{t_f} \Gamma(t) dt \quad (10.25)$$

The state vector is defined as

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{r}(t) \\ \mathbf{v}(t) \end{bmatrix} \quad (10.26)$$

where $\mathbf{r}(t)$ is the spacecraft position vector and $\mathbf{v}(t)$ is its velocity vector. The mass m can be kept track of without defining it to be a state variable by noting that

$$m(t) = m_o e^{-F(t)/c} \quad (10.27)$$

where

$$F(t) = \int_{t_o}^t \Gamma(\xi) d\xi \quad (10.28)$$

We note from Eq. (10.28) that $F(t_f)$ is equal to the cost J_{CSI} . In the constant thrust case Γ varies according to $\dot{\Gamma} = \frac{1}{c}\Gamma^2$, which is consistent with the mass decreasing linearly with time. (See Exercise 1.)

The equation of motion is

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{g}(\mathbf{r}) + \Gamma \mathbf{u} \end{bmatrix} \quad (10.29)$$

with the initial state $\mathbf{x}(t_o)$ specified.

The first-order necessary conditions for an optimal CSI trajectory were first derived by Lawden [1963] using classical calculus of variations. In the derivation that follows, an optimal control theory formulation is used, but the derivations and examples are analogous to those of Lawden and Breakwell. One significant difference is that the mass is not considered a state variable, but is kept track of separately which simplifies the state equations and the adjoint equations by having fewer variables.

In order to minimize the cost in Eq. (10.25) we form the Hamiltonian using Eq. (10.29) as

$$H = \Gamma + \boldsymbol{\lambda}_r^T \mathbf{v} + \boldsymbol{\lambda}_v^T [\mathbf{g}(\mathbf{r}) + \Gamma \mathbf{u}] \quad (10.30)$$

The adjoint (costate) equations are then

$$\dot{\lambda}_r^T = - \frac{\partial H}{\partial \mathbf{r}} = - \lambda_v^T \mathbf{G}(\mathbf{r}) \quad (10.31)$$

$$\dot{\lambda}_v^T = - \frac{\partial H}{\partial \mathbf{v}} = - \lambda_r^T \quad (10.32)$$

where

$$\mathbf{G}(\mathbf{r}) \equiv \frac{\partial \mathbf{g}(\mathbf{r})}{\partial \mathbf{r}} \quad (10.33)$$

is the symmetric 3×3 *gravity gradient matrix*. (See Exercise 2.)

Example 10.1 Derivatives of the gravity gradient matrix.

For the inverse-square gravitational field: $\mathbf{g}(\mathbf{r}) = -(\mu/r^2)\mathbf{r}/r = -(\mu/r^3)\mathbf{r}$, show that the gravity gradient matrix $\mathbf{G}(\mathbf{r})$ of Eq. (10.33) is equal to $\mathbf{G}(\mathbf{r}) = \frac{\mu}{r^5}(3\mathbf{r}\mathbf{r}^T - r^2\mathbf{I}_3)$, where $\mathbf{I}_3$ is the 3×3 identity matrix.

For $g = -\frac{\mu r}{r^3}$ we have:

$$\mathbf{G} = \frac{\partial \mathbf{g}}{\partial \mathbf{r}} = \left[r^3(-\mu \frac{\partial \mathbf{r}}{\partial \mathbf{r}}) + \mu \mathbf{r}(3r^2 \frac{\partial r}{\partial \mathbf{r}}) \right] / r^6 \quad (10.34)$$

Using $\partial \mathbf{r} / \partial \mathbf{r} = \mathbf{I}_3$ and differentiating $r^2 = \mathbf{r}^T \mathbf{r}$ yields $2r(\partial r / \partial \mathbf{r}) = 2\mathbf{r}^T$, so $\partial r / \partial \mathbf{r} = \mathbf{r}^T / r$ and we obtain:

$$\mathbf{G} = \frac{3\mu r^2 \mathbf{r} \mathbf{r}^T / r - \mu r^3 \mathbf{I}_3}{r^6} = \frac{\mu}{r^5} (3\mathbf{r} \mathbf{r}^T - r^2 \mathbf{I}_3) \quad (10.35)$$

Following the notation of section 3.3.1 the terminal constraints for optimal rocket trajectories are of the form

$$\Psi[t_f, \mathbf{r}(t_f), \mathbf{v}(t_f)] = \mathbf{0} \quad (10.36)$$

which may describe an orbital intercept, rendezvous, etc. The boundary conditions on Eqs. (10.31) and (10.32) are given in terms of

$$\Phi \equiv \boldsymbol{\nu}^T \Psi[\mathbf{r}(t_f), \mathbf{v}(t_f), t_f] \quad (10.37)$$

as

$$\lambda_r^T(t_f) = \frac{\partial \Phi}{\partial \mathbf{r}(t_f)} = \boldsymbol{\nu}^T \frac{\partial \Psi}{\partial \mathbf{r}(t_f)} \quad (10.38)$$

$$\lambda_v^T(t_f) = \frac{\partial \Phi}{\partial \mathbf{v}(t_f)} = \boldsymbol{\nu}^T \frac{\partial \Psi}{\partial \mathbf{v}(t_f)} \quad (10.39)$$

where $\boldsymbol{\nu}$ is a constant Lagrange multiplier vector.

There are two control variables, the thrust direction $\mathbf{u}$ and the thrust acceleration magnitude Γ , that must be chosen to satisfy the Minimum Principle, i.e., to minimize the instantaneous value of the Hamiltonian H , as discussed in chapter 6. By inspection, the Hamiltonian of Eq. (10.30) is minimized over the choice of thrust direction by aligning the unit vector $\mathbf{u}(t)$ opposite to the adjoint vector $\boldsymbol{\lambda}_v(t)$. Because of the significance of the vector $-\boldsymbol{\lambda}_v(t)$ Lawden [1963] termed it the *primer vector* $\mathbf{p}(t)$:

$$\mathbf{p}(t) \equiv -\boldsymbol{\lambda}_v(t) \quad (10.40)$$

The optimal thrust unit vector is then in the direction of the primer vector, specifically:

$$\mathbf{u}(t) = \frac{\mathbf{p}(t)}{p(t)} \quad (10.41)$$

and

$$\boldsymbol{\lambda}_v^T \mathbf{u} = -\lambda_v = -p \quad (10.42)$$

in the Hamiltonian of Eq. (10.30).

From Eqs. (10.32) and (10.38) we see that

$$\dot{\mathbf{p}}(t) = \boldsymbol{\lambda}_r(t) \quad (10.43)$$

Equations (10.31), (10.32), (10.38), and (10.41) combine to yield the *primer vector equation*

$$\ddot{\mathbf{p}} = \mathbf{G}(\mathbf{r})\mathbf{p} \quad (10.44)$$

The boundary conditions on the solution to Eq. (10.42) are obtained from Eqs. (10.36) and (10.37):

$$\mathbf{p}^T(t_f) = -\boldsymbol{\nu}^T \frac{\partial \Psi}{\partial \mathbf{v}(t_f)} \quad (10.45)$$

$$\dot{\mathbf{p}}^T(t_f) = \boldsymbol{\nu}^T \frac{\partial \Psi}{\partial \mathbf{r}(t_f)} \quad (10.46)$$

We note that in Eq. (10.43) the final value of the primer vector for an optimal intercept (only final position specified) is the zero vector, because the terminal constraint Ψ does not depend on the final velocity $\mathbf{v}(t_f)$.

Using Eqs. (10.38) and (10.41) the Hamiltonian of Eq. (10.30) can be rewritten as

$$H = -(p-1)\Gamma + \dot{\mathbf{p}}^T \mathbf{v} - \mathbf{p}^T \mathbf{g} \quad (10.47)$$

To minimize the Hamiltonian over the choice of the thrust acceleration magnitude, Γ , we note that the Hamiltonian is a linear function of Γ , and thus the minimizing value for

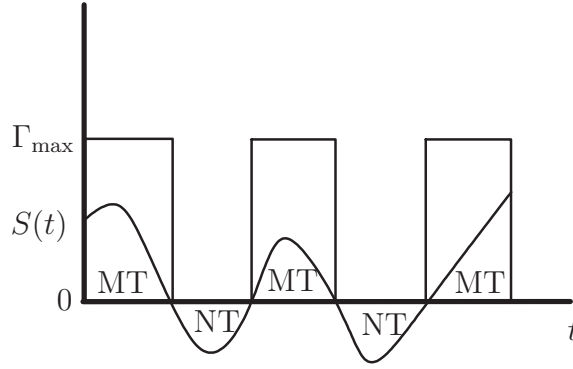


Figure 10.1: Three-burn CSI switching function and thrust profile.

$0 \leq \Gamma \leq \Gamma_{max}$ will depend on the algebraic sign of the coefficient of Γ in Eq. (10.45). It is convenient to define the *switching function*

$$S(t) \equiv p(t) - 1 \quad (10.48)$$

The choice of the thrust acceleration magnitude, Γ , that minimizes H is then given by the bang-bang control law:

$$\Gamma = \begin{cases} \Gamma_{max} & \text{for } S > 0 \quad (p > 1) \\ 0 & \text{for } S < 0 \quad (p < 1) \end{cases} \quad (10.49)$$

That is, the thrust magnitude switches between its limiting values of 0 (an NT, null-thrust, arc) and T_{max} (an MT, maximum-thrust, arc) each time $S(t)$ passes through 0 [i.e. $p(t)$ passes through 1] according to Eq. (10.47). Figure 10.1 shows an example switching function for an NT-MT-NT sequence.

The possibility also exists that $S(t) \equiv 0$ [$p(t) \equiv 1$] on an interval of finite duration. From Eq. (10.45) it is evident that in this case the thrust acceleration magnitude is not determined by the Minimum Principle and may take on intermediate values between 0 and Γ_{max} . This IT (intermediate thrust arc) in Lawden [1963] is called a *singular arc* in optimal control.

From the Weierstrass-Erdmann corner conditions of chapter 8, we know that, at a corner separating thrust arcs of different types, λ_r and λ_v , and hence $\mathbf{p}$ and $\dot{\mathbf{p}}$, are continuous. Equation (10.46) then indicates that the switching function $S(t)$ is also continuous.

Lawden explained to co-author Prussing the origin of the term *primer vector* in a personal letter in 1990: “In regard to the term ‘primer vector’ you are quite correct in your supposition. I served in the artillery during the war [World War II] and became familiar with the initiation of the burning of cordite by means of a primer charge. Thus, $p=1$ is the signal for the rocket motor to be ignited.”

It follows then from Eq. (10.27) that, if $T = T_{max}$ and the engine is on for a total of Δt time units,

$$\Gamma_{max}(t) = e^{F(t)/c} T_{max}/m_o = T_{max}/(m_o - b_{max}\Delta t) \quad (10.50)$$

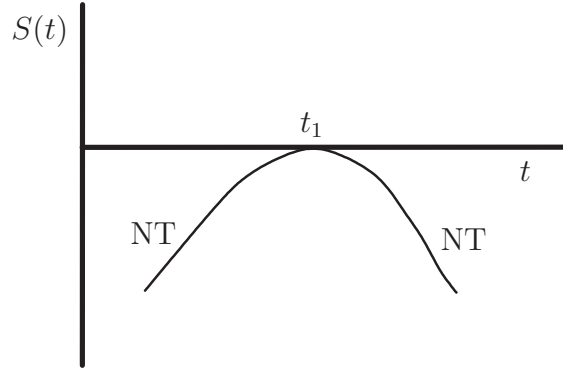


Figure 10.2: Switching function for an impulsive thrust.

Even though the gravitational field is time-invariant, the Hamiltonian in this formulation does not provide a first integral (constant of the motion) on an MT arc, because Γ_{max} is an explicit function of time as shown in Eq. (10.48). On NT and IT arcs, however, the Hamiltonian is constant. (See Example 10.2.) From the corner conditions we note that the Hamiltonian is continuous at a corner between arcs of different types, which is also evident from Eq. (10.45) because $S = 0$ (i.e. $p = 1$) at those instants when Γ is discontinuous.

10.5.2 Optimal Impulsive Trajectory

For a high-thrust CSI engine the thrust durations are very small compared with the times between thrusts. Because of these short durations we can approximate each MT arc as an impulse (a Dirac delta function) having unbounded magnitude ($\Gamma_{max} \rightarrow \infty$) and zero duration. The primer vector then determines both the optimal times and directions of the thrust impulses with $p \leq 1$ corresponding to $S \leq 0$. The impulses can occur only at those instants at which $S = 0$ ($p = 1$). These impulses are separated by NT arcs along which $S < 0$ ($p < 1$). At the impulse times, the primer vector is a unit vector in the optimal thrust direction. Figure 10.2 shows a switching function for an impulsive thrust at time t_1 .

The primer vector $\mathbf{p}$ is defined at an impulse time and $\mathbf{r}$ is continuous at an impulse (recall that it is $\dot{\mathbf{r}} = \mathbf{v}$ that is discontinuous). Combined with the primer vector Eq. (10.42) it follows that $\mathbf{p}$, $\dot{\mathbf{p}}$, and $\ddot{\mathbf{p}}$ are continuous at an impulse.

Figure 10.3 shows a hypothetical cusp for which $\dot{S} \neq 0$ when $S = 0$. However, as we will show, a cusp is not possible. From the Hamiltonian of Eq. (10.45) and the Weierstrass-Erdmann corner condition

$$H^+ - H^- = 0 = \dot{\mathbf{p}}^T(\mathbf{v}^+ - \mathbf{v}^-) = \dot{\mathbf{p}}^T \Delta \mathbf{v} = \Delta \mathbf{v} \dot{\mathbf{p}}^T \mathbf{p} \quad (10.51)$$

because $\mathbf{p}$, $\dot{\mathbf{p}}$, and $\mathbf{g}$ are continuous, $S = 0$ ($p = 1$), and $\Delta \mathbf{v} = \Delta \mathbf{v} \mathbf{p}$. Thus $\dot{\mathbf{p}}^T \mathbf{p} = \dot{p} = 0$ (See Exercise 7), which implies that $\dot{S} = 0$. So the hypothetical case shown in Fig. 10.3 does not exist and there can be no cusp. This argument does not apply at the terminals of the

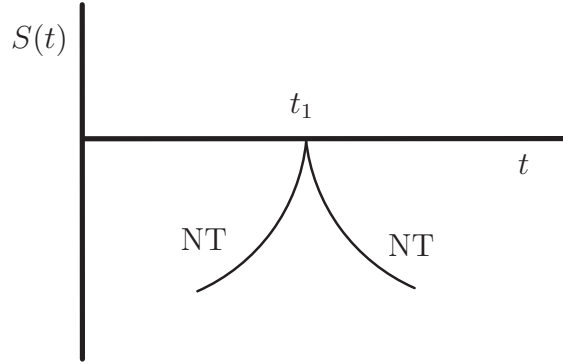


Figure 10.3: Hypothetical cusp for which $\dot{S} \neq 0$ when $S = 0$.

trajectory because the corner conditions do not apply at the terminals. So in general $\dot{p} \neq 0$ at the terminals.

In Lion and Handelsman [1968] a procedure is developed to iteratively improve a nonoptimal trajectory (that violates one or more of the necessary conditions summarized below) to converge to an optimal trajectory. As discussed in Prussing [2010], when adding a midcourse impulse lowers the cost, the cost gradients with respect to the midcourse impulse position and time (which are used to iterate on these variables) depend on the discontinuities in $\dot{\mathbf{p}}$ and H at the midcourse impulse. As the midcourse impulse position and time approach their optimal values, these discontinuities tend to zero.

The necessary conditions (NC) for an optimal impulsive trajectory, first derived by Lawden [1963], are as follows

1. The primer vector and its first derivative are continuous everywhere.
2. The magnitude of the primer vector satisfies $p(t) \leq 1$ with the impulses occurring at those instants at which $p = 1$.
3. At the impulse times the primer vector is a unit vector in the optimal thrust direction.
4. As a consequence of the above conditions, $dp/dt = \dot{p} = \dot{\mathbf{p}}^T \mathbf{p} = 0$ at an intermediate impulse (not at the initial or final time). (See Exercise 7.)

For a linear system, Prussing [1995] shows that these NC are also sufficient conditions (SC) for an optimal trajectory. Prussing also derives an upper bound on the number of impulses required for an optimal solution. However, for a nonlinear system no upper bound exists.

We note also that for a thrust impulse at time t_k

$$\Gamma(t) = \Delta \mathbf{v}_k \delta(t - t_k) \quad (10.52)$$

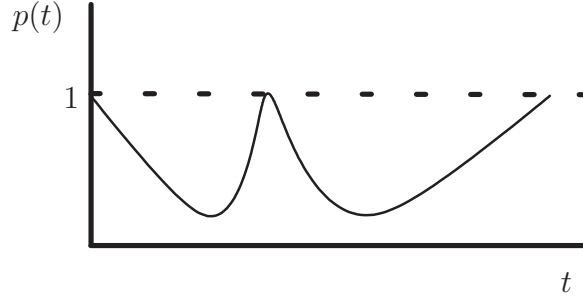


Figure 10.4: Sample primer vector history for a three-impulse trajectory. The magnitude of the primer vector satisfies $p(t) \leq 1$ with the impulses occurring at those instants at which $p = 1$.

and, from Eq. (10.28), the $\Delta \mathbf{v}_k$ can be expressed as

$$\Delta \mathbf{v}_k = \int_{t_k^-}^{t_k^+} \Gamma(t) dt = F(t_k^+) - F(t_k^-) \quad (10.53)$$

where t_k^+ and t_k^- are times immediately after and before the impulse time, respectively. Equation (10.27) then becomes the familiar solution to the rocket equation, Eq. (10.9), with the interval term equal to zero.

$$m(t_k^+) = m(t_k^-) e^{-\Delta \mathbf{v}_k / c} \quad (10.54)$$

Figure 10.4 illustrates the primer vector magnitude for a three-impulse trajectory.

10.5.3 Optimal Variable Specific Impulse Trajectory

A variable specific impulse (VSI) engine is also known as a power-limited (PL) engine, because the power source is separate from the engine itself, e.g., solar panels, radioisotope thermoelectric generator, etc. The power delivered to the engine is bounded between 0 and a maximum value P_{max} , with the optimal value being constant and equal to the maximum, as discussed after Eq. (10.20). The cost functional representing minimum-propellant consumption for the VSI case is given by Eq. (10.22) as

$$J = \frac{1}{2} \int_{t_o}^{t_f} \Gamma^2(t) dt \quad (10.55)$$

Writing Γ^2 as $\mathbf{\Gamma}^T \mathbf{\Gamma}$ we obtain the corresponding Hamiltonian function:

$$H = \frac{1}{2} \mathbf{\Gamma}^T \mathbf{\Gamma} + \boldsymbol{\lambda}_r^T \mathbf{v} + \boldsymbol{\lambda}_v^T [\mathbf{g}(\mathbf{r}) + \mathbf{\Gamma}] \quad (10.56)$$

For the VSI case there is no need to consider the thrust acceleration magnitude and direction separately, so the vector $\mathbf{\Gamma}$ is used in place of the term $\Gamma \mathbf{u}$ that appears in Eq. (10.30).

Because H is a nonlinear function of $\mathbf{\Gamma}$, the Minimum Principle is applied by setting

$$\frac{\partial H}{\partial \mathbf{\Gamma}} = \mathbf{\Gamma}^T + \boldsymbol{\lambda}_v^T = \mathbf{0}^T \quad (10.57)$$

or

$$\mathbf{\Gamma}(t) = -\boldsymbol{\lambda}_v(t) = \mathbf{p}(t) \quad (10.58)$$

using the definition of the primer vector in Eq. (10.38). Thus for a VSI engine the optimal thrust acceleration vector is equal to the primer vector: $\mathbf{\Gamma}(t) = \mathbf{p}(t)$. Now Eq. (10.1) $\ddot{\mathbf{r}} = \mathbf{g}(\mathbf{r}) + \mathbf{\Gamma}$, can be combined with Eq. (10.42) to yield a fourth-order differential equation in $\mathbf{r}$:

$$\mathbf{r}^{iv} - \dot{\mathbf{G}}\dot{\mathbf{r}} + \mathbf{G}(\mathbf{g} - 2\ddot{\mathbf{r}}) = \mathbf{0} \quad (10.59)$$

Every solution to Eq. (10.57) is an optimal VSI trajectory through the gravity field $\mathbf{g}(\mathbf{r})$. However, desired boundary conditions, such as specified position and velocity vectors at the initial and final times must be satisfied.

We also note that from Eq. (10.55):

$$\frac{\partial^2 H}{\partial \mathbf{\Gamma}^2} = \frac{\partial}{\partial \mathbf{\Gamma}} \left(\frac{\partial H}{\partial \mathbf{\Gamma}} \right)^T = \mathbf{I}_3 \quad (10.60)$$

where $\mathbf{I}_3$ is the 3×3 identity matrix. Equation (10.58) shows that the (Hessian) matrix of second partial derivatives is positive definite, verifying that H is minimized.

Because the optimal VSI thrust acceleration is continuous, the procedure in Prussing and Sandrik [2005] to test whether second-order NC and SC are satisfied can be applied. Equation (10.57) shows that an NC for minimum cost (Hessian matrix positive semidefinite) and part of the SC (Hessian matrix positive definite) are satisfied. The other condition that is both an NC and SC is the Jacobi no-conjugate-point condition discussed in section 6.6. Prussing and Sandrik [2005] provide the details of the no-conjugate-point test.

Example 10.2 The Hamiltonian for optimal rocket trajectories.

1. For an optimal CSI trajectory show that the Hamiltonian in Eq. (10.45) is constant on IT and NT arcs for a static gravitational field. Hint: calculate $\dot{H}$.
2. For an optimal VSI trajectory show that the Hamiltonian in Eq. (10.54) is constant.

Solution:

1. On an IT arc we have $p = 1$ and on an NT arc $\Gamma = 0$, so the first term in Eq. (10.45) is zero in both cases. Then $\dot{H} = \ddot{\mathbf{p}}^T \mathbf{v} + \dot{\mathbf{p}}^T \dot{\mathbf{v}} - \dot{\mathbf{p}}^T \mathbf{g} - \mathbf{p}^T \dot{\mathbf{g}}$. Using $\ddot{\mathbf{p}} = \mathbf{G}\mathbf{p}$, $\dot{\mathbf{v}} = \mathbf{g} + \mathbf{\Gamma}$ and $\dot{\mathbf{g}} = \mathbf{G}\mathbf{v}$ (See Exercise 3.), $\dot{H} = \mathbf{p}^T \mathbf{G}\mathbf{v} + \dot{\mathbf{p}}^T (\mathbf{g} + \mathbf{\Gamma}) - \dot{\mathbf{p}}^T \mathbf{g} - \mathbf{p}^T \mathbf{G}\mathbf{v} = \dot{\mathbf{p}}^T \mathbf{\Gamma}$. On an NT arc $\mathbf{\Gamma} = \mathbf{0}$ and on IT arc $p = 1$, $\mathbf{\Gamma} = \Gamma \mathbf{p}$, and $\dot{\mathbf{p}}^T \mathbf{p} = \dot{p} = 0$.
2. Substituting $\mathbf{\Gamma} = \mathbf{p}$, $\boldsymbol{\lambda}_r = \dot{\mathbf{p}}$, and $\boldsymbol{\lambda}_v = -\mathbf{p}$ into the expression for $\dot{H}$ we have $\dot{H} = \mathbf{p}^T \dot{\mathbf{p}} + \ddot{\mathbf{p}}^T \mathbf{v} + \dot{\mathbf{p}}^T (\mathbf{g} + \mathbf{p}) - \dot{\mathbf{p}}^T (\mathbf{g} + \mathbf{p}) - \mathbf{p}^T (\mathbf{G}\mathbf{v} + \dot{\mathbf{p}})$. Using solution (a) and the fact that $\mathbf{G}$ is symmetric we find that the Hamiltonian is constant.

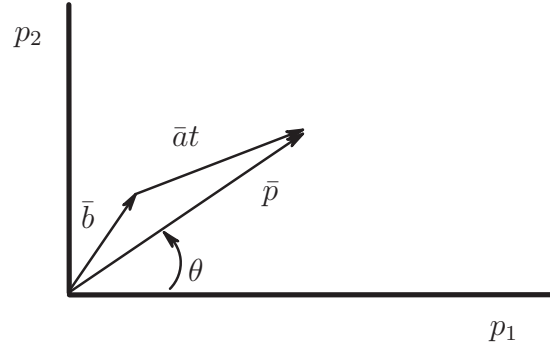


Figure 10.5: In a uniform gravity field the primer vector is restricted to a plane which means that the thrust direction may only vary in a plane.

10.6 Optimal Trajectories in a Uniform Field

For a uniform gravitational field, we have $\mathbf{g} = \text{constant}$ which implies from Eqs. (10.33) and (10.44) that $\dot{\mathbf{p}} = 0$. Thus, the general solution for the primer vector is

$$\mathbf{p} = \mathbf{a}t + \mathbf{b} \quad (10.61)$$

where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors. Equation (10.61) is, obviously, the equation of a straight line. In the special case where $\mathbf{a}=0$, the thrust direction never varies throughout the maneuver. In the general case, the thrust direction may vary in a plane, determined by the vectors $\mathbf{a}$ and $\mathbf{b}$, as illustrated in Fig. 10.5. Components of the primer vector (in general) are of the form:

$$p_1 = a_1t + b_1 \quad (10.62a)$$

$$p_2 = a_2t + b_2 \quad (10.62b)$$

$$p_3 = 0 \quad (10.62c)$$

where we have chosen a convenient set of coordinates so that $\mathbf{p}$ is in the xy plane.

The angle made by the primer vector (which provides us with the thrust direction) is given by

$$\tan \theta = \frac{p_2}{p_1} = \frac{a_2t + b_2}{a_1t + b_1} \quad (10.63)$$

which we recognize as the bilinear tangent steering law encountered in chapter 4. [See Eq. (4.61)].

The optimal trajectory in a uniform field can include an IT arc only in special circumstances.

Lawden [1963] shows (for CSI) that for an IT arc $\mathbf{a}=0$ and $\mathbf{p} = \mathbf{b}$. Thus the thrust direction is constant throughout the maneuver. Under these conditions the equations of motion can be integrated, leading to highly constrained end conditions which rule out most trajectories, in which case no IT arcs are optimal. On the other hand, if these end conditions

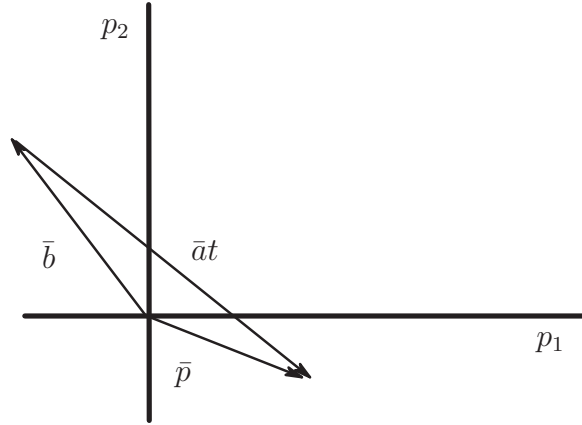


Figure 10.6: The primer vector in a uniform field may decrease in magnitude initially, reach a minimum, and then increase.

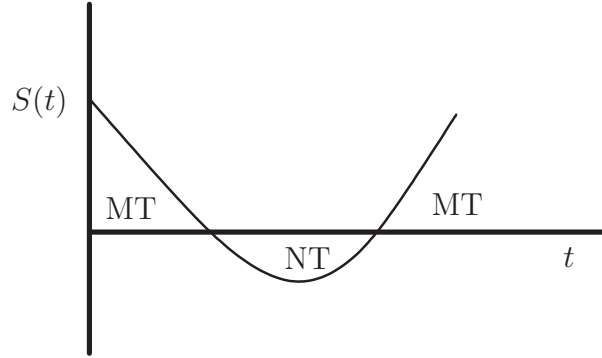


Figure 10.7: At most only three subarcs can exist MT, NT, MT according to the switching function for the primer vector in a uniform field.

are satisfied, it is possible to have a number of IT arcs. In the case where the end conditions rule out an IT arc, the optimal trajectory will consist of no more than three NT and MT subarcs (as shown by Leitmann [1959]). In Fig. 10.6 we illustrate possible behavior of the primer vector. In the figure we note that p is the distance from the origin. In the general case, we can have three phases:

1. The magnitude of the primer vector, p , starts out at some initial value,
2. p decreases to a minimum (or to zero), then
3. p increases to the final value.

For cost functional, J_{CSI} , if the initial value of p is greater than unity, the minimum less than unity, and the final greater than unity, then from Eq. (10.46) the switching function will have values of $S > 1$, $S < 1$, and $S > 1$, respectively corresponding to Eq. (10.47) and Fig. 10.7. We conclude that we can have at most three subarcs: MT, NT, MT in that

order. We also note that for impulsive thrusts there are at most two impulses. These simple examples demonstrate the power of Lawden's primer vector theory.

10.7 Summary

For optimal rocket trajectories in space, we assume that the only forces acting on the spacecraft are due to gravity and rocket thrust. The thrusters can be pointed in arbitrary directions and the magnitude of the thrust may be limited. The cost to be minimized, J , is the very important one of minimum-propellant consumption (which is equivalent to maximizing final mass). For a space mission, every kilogram of propellant saved by optimization provides an additional kilogram of payload at the final destination.

For this problem, Lawden found, using the calculus of variations, that the adjoint (costate) vector corresponding to the velocity vector provides the optimal thrust direction. Analysis of this vector, the *primer vector*, indicates when null thrust (NT), maximum thrust (MT), or intermediate thrust (IT) arcs are optimal for a constant specific impulse (CSI) engine. For a variable specific impulse (VSI) engine the magnitude of the optimal thrust acceleration is equal to the magnitude of the primer vector.

If the thrust magnitude in the CSI case is unbounded, we have the impulsive thrust case and the change in velocity is instantaneous. In reality, impulsive thrust does not exist, but the short duration of deep space thrust maneuvers compared with the time between maneuvers can be modeled as impulsive.

Lawden neatly summarizes the necessary conditions for optimal impulsive thrust trajectories in four statements involving the primer vector. These conditions also form the basis for primer vector theory, in which the primer vector evaluated on a nonoptimal impulsive trajectory can be used to determine an optimal trajectory for the same terminal conditions.

When Lawden's primer vector is applied to the problem of optimal trajectories in a uniform gravitational field with bounded thrust, we rediscover the bilinear tangent steering law from our launch from flat-Earth problem.

10.8 Exercises

1. Show that for a constant thrust, $\dot{\Gamma} = \frac{1}{c}\Gamma^2$ corresponds to a constant mass flow rate b . Hint: $m(t) = m_o - bt$.
2. Show that the gravity gradient matrix in Eq. (10.33) is symmetric for a conservative gravitational field. Hint: write $g_i = -\frac{\partial V}{\partial r_i}$, where V is a (scalar) gravitational potential function, then calculate $G_{ij} = \frac{\partial g_i}{\partial r_j}$.
3. Assuming a static gravitational field where $\mathbf{g}(\mathbf{r})$ is not an explicit function of time, determine under what circumstances $\dot{\mathbf{g}} \equiv \frac{d\mathbf{g}}{dt}$ is nonzero.
4. Consider the gravity gradient matrix for the case of a central gravitational field:

- 4a. Determine the gravity gradient matrix $\mathbf{G}(\mathbf{r})$ for the general central gravitational field: $\mathbf{g}(\mathbf{r}) = g(r)\mathbf{r}$. Hint: the answer depends on the variables $\mathbf{r}$, $g(r)$, and $g'(r) \equiv dg/dr$.
- 4b. Verify that your answer specializes to the result in Example 10.1. for $g(r) = -\frac{\mu}{r^3}$.
5. Consider the vector $\mathbf{A} = \mathbf{p} \times \mathbf{v} - \dot{\mathbf{p}} \times \mathbf{r}$ referenced by Marec [1979]:
 - 5a. Determine whether $\mathbf{A}$ is constant on an optimal continuous-thrust trajectory in the general central gravitational field of Exercise 4a. Discuss both the CSI and VSI cases, including the discontinuous thrust in the CSI case.
 - 5b. Determine whether the vector $\mathbf{A}$ is continuous at an optimal impulsive thrust.
6. Consider an optimal impulsive trajectory in an inverse-square gravitational field:
 - 6a. Show that the scalar a defined by Lion and Handelsman [1968] is constant on an NT arc between impulses, where $a = 2\dot{\mathbf{p}}^T \mathbf{r} + \mathbf{p}^T \mathbf{v} - 3Ht$.
 - 6b. Demonstrate that the discontinuity in the variable a at an optimal impulse is equal to the magnitude of the velocity change $\Delta \mathbf{v}$.
 - 6c. Using the result of 6b. determine a new quantity $\hat{a}$ that is both constant between impulses and continuous at an optimal impulse. Hint: form $\hat{a}$ by adding a term to a that depends on the mass.
7. Show that when $p = 1$, $\dot{p} = \dot{\mathbf{p}}^T \mathbf{p}$, where $\dot{p} \equiv dp/dt \neq |\dot{\mathbf{p}}|$ in general. Hint: differentiate $p^2 = \mathbf{p}^T \mathbf{p}$.
8. Consider the optimal VSI trajectory equation:
 - 8a. Derive Eq. (10.59).
 - 8b. Specialize your result for a uniform gravitational field, where $\mathbf{g}(\mathbf{r})$ is a constant vector.
 - 8c. Write the general solution $\mathbf{r}(t)$ to the equation obtained in 8b.
 - 8d. Based on 8c. write an expression for the optimal thrust acceleration $\mathbf{\Gamma}(t)$.
9. Consider the problem of optimal intercept:
 - 9a. Based on the sentence following Eq. (10.46), specialize the form of the primer vector in Eq. (10.59) for an optimal intercept.
 - 9b. Qualitatively describe the behavior of the thrust direction.
 - 9c. Describe the possible subarc sequences involving MT, NT, and IT arcs.
 - 9d. For impulsive thrusts, what is the maximum number of impulses?

10.9 True or False Quiz for Chapters 6–10

Answer the following questions in the context of the material presented in chapters 6–10.

1. According to the general theory of optimal rocket trajectories (and assuming $p \neq 0$) whenever the thruster is operating, the thrust must act in the direction of the primer vector, $\mathbf{p}$.

True

False

2. In the theory of optimal rocket trajectories, when impulsive thrusts are considered, then the switching function, $S = p - 1$, is never greater than zero.

True

False

3. Derivation of the rules for the primer vector is based mainly on the Minimum Principle and the Weierstrass-Erdmann corner conditions.

True

False

4. In the derivation of the Weierstrass-Erdmann corner conditions, it is shown that $\lambda(t)$ and $H(t)$ are continuous at corners.

True

False

5. The Generalized Legendre-Clebsch condition was shown to be directly derivable from the Minimum Principle.

True

False

6. If $T_{\max} \rightarrow \infty$ then all optimal maneuvers must be impulsive.

True

False

7. In primer theory we always have $\mathbf{p} = \mathbf{v}$ and $\dot{\mathbf{p}} = \mathbf{g}$.

True

False

8. No modifications of the Minimum Principle are required for the State Variable Inequality Constraint (SVIC) Problem.

True

False

9. If $x^*(t)$ is a weak extremal, then it may be possible to find a control which provides a lower cost.

True

False

10. For a uniform gravity field, if the end conditions rule out an IT arc, then the optimal trajectory will consist of not more than three NT and MT subarcs.

True

False

11. The Minimum Principle always tells us everything we need to know to find optimal trajectories.

True

False

12. The main difference between the Pontryagin Minimum Principle and the Weierstrass condition is the class of functions of the admissible controls.

True

False

13. In general, the Legendre-Clebsch condition applies to a broader class of controls than does the Pontryagin Minimum Principle.

True

False

14. If $x^*(t)$ is a weak extremal, then it is also a strong extremal.

True

False

15. On a corner $x(t_c^-) \neq x(t_c^+)$.

True

False

Solution: 1T, 2T, 3T, 4T, 5F, 6F, 7F, 8F, 9T, 10T, 11F, 12T, 13F, 14F, 15F.

10.10 References

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